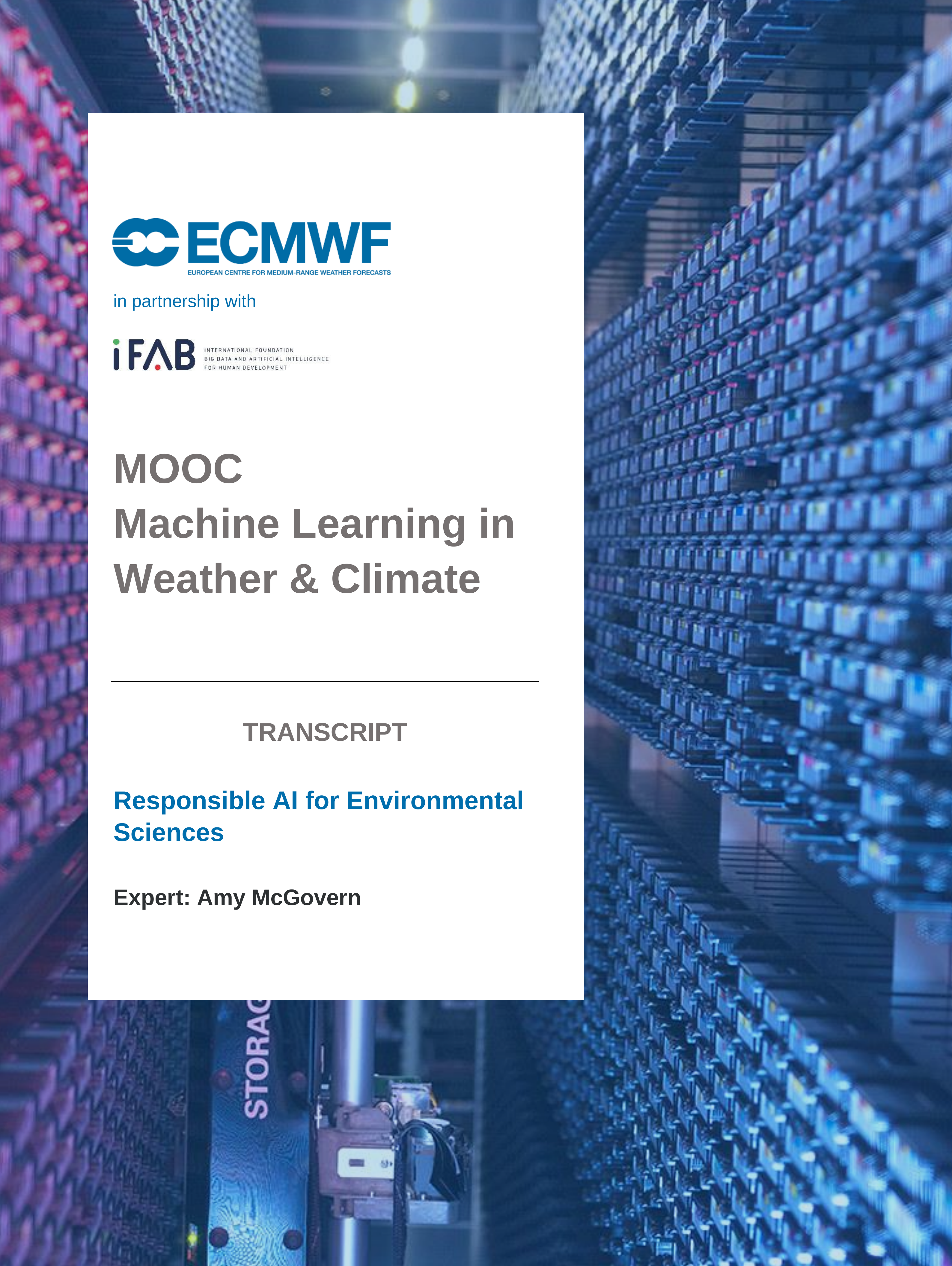

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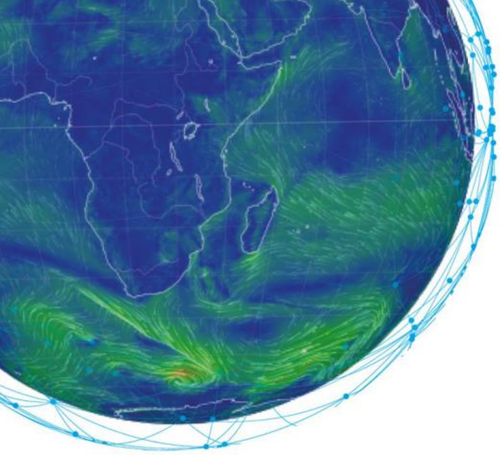
MOOC
Machine Learning in Weather & Climate

TRANSCRIPT

Responsible AI for Environmental Sciences

Expert: Amy McGovern

University of Luxembourg
Competence Centre



Responsible AI for Environmental Sciences

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Introduction

Hello and welcome. We are here today to talk about why we need to ensure that the AI methods we are developing for weather and climate need to be developed in an ethical and responsible manner. With us today is Professor Amy McGovern, a leading expert in the development of AI for weather and climate applications. Prof McGovern is a professor in both Computer Science and Meteorology at the University of Oklahoma, in the United States. Professor McGovern also leads the AI Institute for Research on Trustworthy AI in Weather, Climate, and Coastal Oceanography, which is funded by the US National Science Foundation. You can learn more about their work on their website at ai2es.org

Lisa: Let's get started with a simple question. We've read about unethical uses of AI in applications outside weather but they seem pretty far afield from weather. What does mis-recognizing the race or gender of a face have to do with weather?

Amy: That's a good question! As you no doubt know by this point in your course, AI models require a lot of high-quality training data to train a high-quality model. Many people think that weather data should be objective and free from the biases that exist with data such as the facial recognition models that you mentioned. In those cases, if a model is trained entirely on white faces, for example, it will be unable to recognize faces of other races because it has never seen them. With weather, we can unintentionally create AI models that reinforce existing biases in our data. These biases could reinforce existing environmental injustice issues also.

Let me give you a couple of examples to help make this more concrete. Let's imagine we want to train an air-quality prediction system that can be used to guide people towards staying indoors on bad air-quality days. Because you want to make this model very precise and personalized to the user, you need fine-scale data to train on. You look online for existing networks of fine-scale air-quality data and you find that such databases exist, as collected from networks of home weather stations. You train your AI model up and it looks to work wonderfully and you are ready to deploy it. What could go wrong with this scenario? Well, think about where your data came from: home stations of air-quality sensors.

Such stations are much more likely to be available in the more affluent areas, meaning you may have unintentionally trained your data to only predict in affluent areas and to ignore the less-affluent areas because you never had any sensor readings there. This unintentionally creates a new AI model that perpetuates environmental injustices!

Let’s look at a second example. In the United States, we have a national network of weather radars. These are used extensively to help in severe weather, for example, in predicting tornadoes and hail. However, a close look at the radar network in the southeast US shows that the radars do not cover many areas with a predominantly black population. This wasn’t done on purpose – it has to do with the expense of the radars and the geography or terrain where they are placed and the goal to cover large cities. But it could mean you train a model that does not predict severe life-altering events such as tornadoes or hail in areas with predominantly black populations. Does this apply in Europe also? Given that there is no EU-wide network of radars, I imagine similar issues could exist across countries. Some countries have good radar coverage and some do not. An algorithm that relies on good radar coverage is not useful in countries without such coverage.

Lisa: The issues you are talking about have to do with biases in the underlying data. If we just solve the data issues so they are free of bias, AI development can proceed without any issues, right?

Amy: Not quite! AI models themselves can learn something you do not intend and thus might generalize or create predictions that make no physical sense! For example, assume you gave an AI model perfect examples of every storm capable of producing a tornado. You knew precisely whether it produced one or not, and if it did, exactly where it went and how strong it was. By the way, this is pretty impossible right now with low-end tornadoes that might not actually cause property damage. If the tornado forms and dissipates all in a remote area, it might not ever be noticed! But assuming we had perfect data, we train our model and we try to see how well it works on our test data. What we don’t know is that somehow it has learned something that really has no grounding in reality. The AI model does not know the laws of physics and somehow it has discovered an idiosyncrasy in the data that does not actually correspond to the laws of physics. But you don't know that unless you really dig into the model. We want to urge you to do that – don’t just take the good predictions for granted. Dig into the model and really understand what it has learned, use the explainable AI techniques that I’m sure you talked about in this course and try to figure out what is going inside the AI model.

Lisa: We are running out of time so let's end with a quick question: You gave us some examples of how AI can go wrong when applied to weather and climate predictions. How do we mitigate this? What do we do to ensure we become developers of ethical and responsible AI?

Amy: This is an excellent question to end on! I hope in our short time I have convinced you that simply developing and deploying AI for weather without thinking about ways in which it could go wrong could potentially create biased AI models. We at AI2ES are in the process of developing guidelines for AI developers for weather and climate applications as well as developing a set of examples of how to mitigate biases in the data. I want to share a few best practices: First, make sure you involve your potential end-users in your development from the start. They are the ones who will be affected by any model issues and they are also the ones who probably recognize data biases easily. Second, make sure you check your data thoroughly for any biases in the training data so you can see how they may appear in the deployed model. Hopefully you can mitigate the effects of the biases but you can't do that or even check if it is working if you don't know they are there! Third, document those biases in your model and be prepared to share your training data and your model and your documentation of your model and data in the open-source community. This can help identify any issues you may have missed! Finally, be willing to pull any deployed models if issues crop up. Perhaps you missed something critical in your bias correction. Be ready to admit this as soon as it is discovered, pull the deployed model from use, and address the issue quickly. This can help the public gain trust in your model as they know you are doing your best to ensure it is developed and used ethically and responsibly.

Summary

Thank you Amy for this great discussion about why we need to ensure that the AI methods we are developing for weather and climate need to be developed in an ethical and responsible manner. We have seen that hidden biases in the data can cause biases in the AI models and then unknowingly perpetuate environmental injustices. We also discussed errors that can crop up in the AI models themselves, largely because they are ignorant of the laws of physics. Finally, we discussed ways to mitigate the issues through working with stakeholders and ensuring that models and data are released in an open fashion.

And for those who were listening to us and that would like to discuss this subject in further detail, please use our Forum on Moodle. It will be a pleasure to hear from you.